

Histogram Equalization Using OpenCV(Grayscale & Color Images)

NAME : INIYA E

REG No : 212224230148

```
In [1]: import cv2
        from matplotlib import pyplot as plt
```

```
In [2]: # Load the color image
        image = cv2.imread('parrot.jpg')
```

```
In [3]: # Convert the image to grayscale
        gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

```
In [4]: plt.imshow(gray_image, cmap='gray')
        plt.title('Original Grayscale Image')
        plt.axis('off')
```

```
Out[4]: (-0.5, 1279.5, 852.5, -0.5)
```

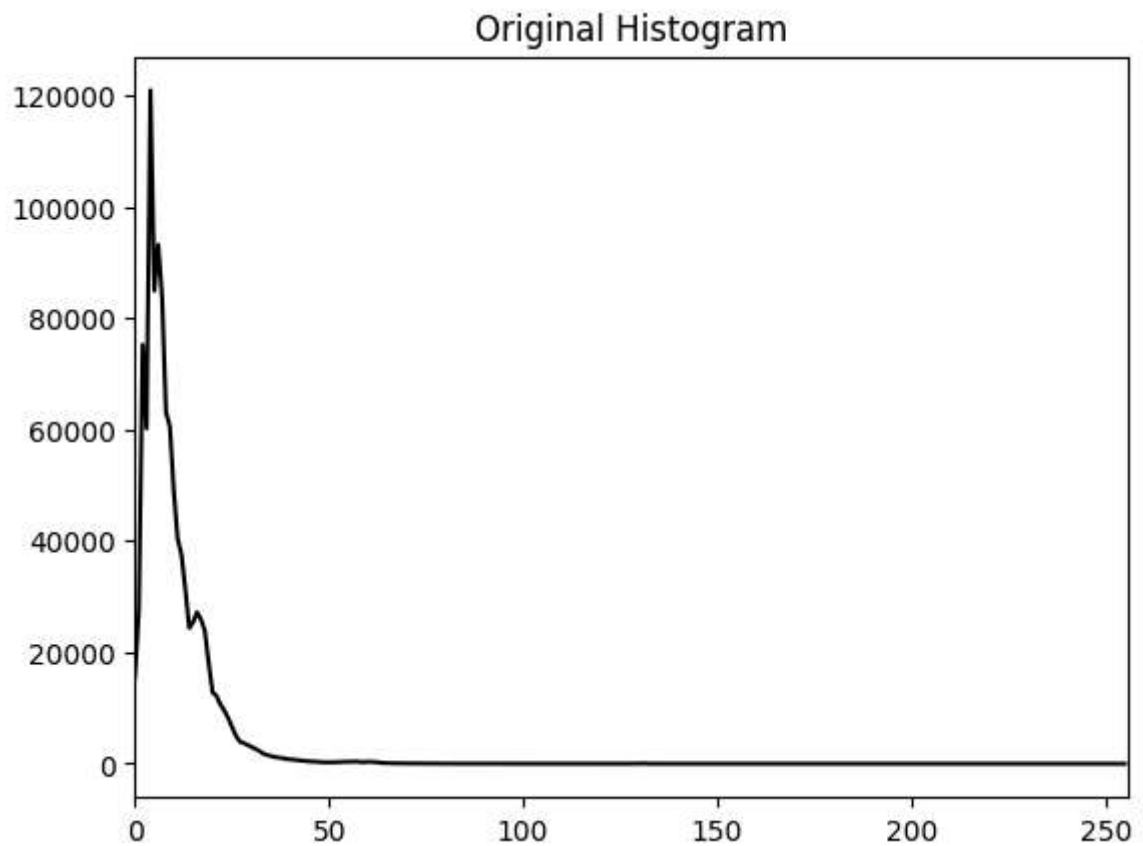
Original Grayscale Image



```
In [5]: hist_original = cv2.calcHist([gray_image], [0], None, [256], [0, 256])
```

```
In [6]: plt.plot(hist_original, color='black')  
plt.title('Original Histogram')  
plt.xlim([0, 256])
```

Out[6]: (0.0, 256.0)



```
In [7]: # Apply histogram equalization  
equalized_image = cv2.equalizeHist(gray_image)
```

```
In [8]: plt.imshow(equalized_image, cmap='gray')  
plt.title('Equalized Image')  
plt.axis('off')
```

```
Out[8]: (-0.5, 1279.5, 852.5, -0.5)
```

Equalized Image



```
In [9]: hist_equalized = cv2.calcHist([equalized_image], [0], None, [256], [0, 256])
```

```
In [10]: plt.plot(hist_equalized, color='black')  
plt.title('Equalized Histogram')  
plt.xlim([0, 256])
```

```
Out[10]: (0.0, 256.0)
```

